Answers learning task 2

2026 – 2027

Exercise 2.122: If $P(A \mid B) = 0.6$, $P(B) = 0.8$, and $P(A) = 0.5$, are the events A and B independent?

Answer

No, because $P(A) = 0.5 \neq P(A \mid B) = 0.6$

Exercise 2.129: The probability that a lab specimen contains high levels of contamination is 0.10. Four samples are checked, and the samples are independent.

(a) What is the probability that none contain high levels of contamination?

(b) What is the probability that exactly one contains high levels of contamination?

(c) What is the probability that at least one contains high levels of contamination?

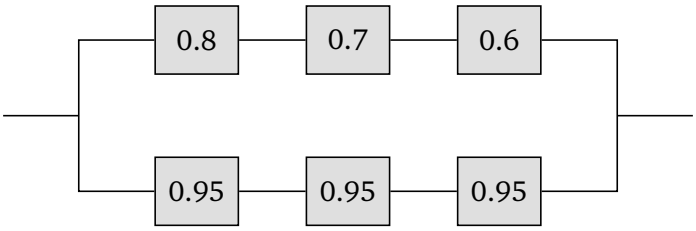
Answer

(a) 0.6561

(b) 0.2916

(c) 0.3439

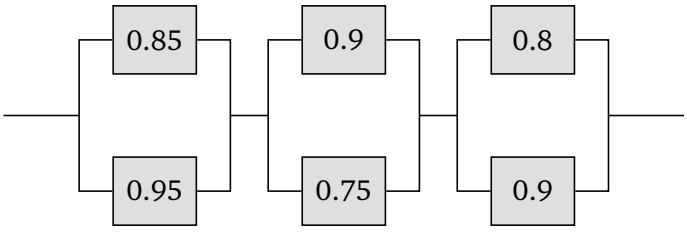
Exercise 2.136: The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?



Answer

0.9053

Exercise 2.137: The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?



Answer

0.9483

Exercise 2.147: Customers are used to evaluate preliminary product designs. In the past, 95 % of highly successful products received good reviews, 60 % of moderately successful products received good reviews, and 10 % of poor products received good reviews. In addition, 45 % of products have been highly successful, 30 % have been moderately successful, and 25 % have been poor products.

(a) What is the probability that a product attains a good review?

(b) If a new design attains a good review, what is the probability that it will be a highly successful product?

(c) If a product does not attain a good review, what is the probability that it will be a highly successful product?

Answer

(a) 0.6325

(b) 0.6759

(c) 0.0612

Exercise 2.149: A new analytical method to detect pollutants in water is being tested. This new method of chemical analysis is important because, if adopted, it could be used to detect three different contaminants — organic pollutants, volatile solvents, and chlorinated compounds — instead of having to use a single test for each pollutant. The makers of the test claim that it can detect high levels of organic pollutants with 99.7 % accuracy, volatile solvents with 99.95 % accuracy, and chlorinated compounds with 89.7 % accuracy. If a pollutant is not present, the test does not signal. Samples are prepared for the calibration of the test and 60 % of them are contaminated with organic pollutants, 27 % with volatile solvents, and 13 % with traces of chlorinated compounds. A test sample is selected randomly.

(a) What is the probability that the test will signal?

(b) If the test signals, what is the probability that chlorinated compounds are present?

Answer

(a) 0.9847

(b) 0.1184

Exercise 2.154: Decide whether a discrete or continuous random variable is the best model for each of the following variables:

(a) The time until a projectile returns to earth.

(b) The number of times a transistor in a computer memory changes state in one operation.

(c) The volume of gasoline that is lost to evaporation during the filling of a gas tank.

(d) The outside diameter of a machined shaft.

(e) The number of cracks exceeding $1.2\,\mathrm{cm}$ in $16\,\mathrm{km}$ of an interstate highway.

(f) The weight of an injection-molded plastic part.

(g) The number of molecules in a sample of gas.

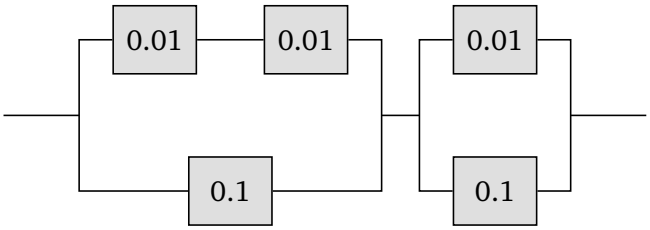
(h) The concentration of output from a reactor.

(i) The current in an electronic circuit.

Answer

1. continuous
2. discrete
3. continuous
4. continuous
5. discrete
6. continuous
7. discrete
8. continuous
9. continuous

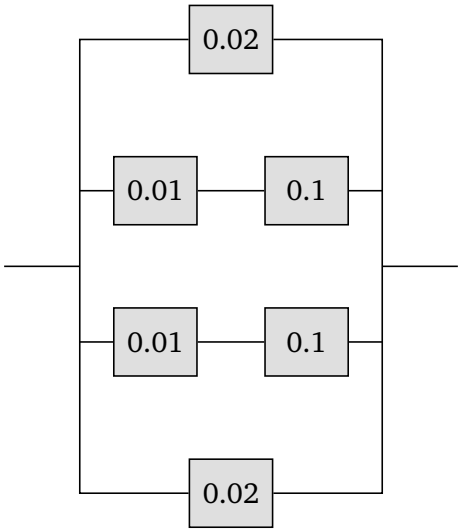
Exercise 2.180: The following circuit operates if and only if there is a path of functional devices from left to right. Assume devices fail independently and that the probability of *failure* of each device is as shown. What is the probability that the circuit operates?



Answer

0.9970

Exercise 2.193: The following circuit operates if and only if there is a path of functional devices from left to right. Assume that devices fail independently and that the probability of *failure* of each device is as shown. What is the probability that the circuit does not operate?



Answer

$4.7524 \cdot 10^{-6}$